

Differentiation

Rewrite as powers of x , bring the power down, subtract one. Then the gradient answers everything: tangents, normals, stationary points.



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THE TOPIC ON ONE MAP

Differentiation

01 Rewrite first

$$\sqrt{x}, \frac{1}{x^2}$$

$$Ax^p + Bx^q$$

02 Using the gradient

gradient at $x = 2$

stationary point

03 Tangents and normals

tangent

normal

$$ax + by + c = 0$$

04 How many stationary points?

$$f'(x) = 0$$

$$b^2 - 4ac$$

05 First principles

chord PQ

$$2 + h$$

THE ONE RULE

$$y = ax^n \implies \frac{dy}{dx} = anx^{n-1}$$

numbers alone disappear · $\frac{dy}{dx}$ = the gradient of the curve = the gradient of the tangent

01 Rewrite First, Then Differentiate

You can only differentiate a sum of powers of x . Brackets, fractions and roots must be rewritten first.

WHEN YOU SEE → YOU DO

$$\sqrt{x}, \sqrt[3]{x}, \frac{1}{x^2}$$

$$\longrightarrow x^{\frac{1}{2}}, x^{\frac{1}{3}}, x^{-2}$$

$$\text{brackets } (2x - 1)^2(x + 1)$$

→ expand first

$$\text{a fraction } \frac{4x^2 + 9}{2\sqrt{x}}$$

→ split: each term ÷ the bottom

$$\text{"in the form } Ax^p + Bx^q\text{"}$$

→ rewrite, then read A, p, B, q

$$\frac{d^2y}{dx^2} / f''(x)$$

→ differentiate twice

WORKED MODEL • A FRACTION WITH A ROOT

$$y = \frac{4x^2 + 9}{2\sqrt{x}}$$

$$1 \quad y = 2x^{\frac{3}{2}} + \frac{9}{2}x^{-\frac{1}{2}}$$

split and rewrite

$$2 \quad \frac{dy}{dx} = 3x^{\frac{1}{2}} - \frac{9}{4}x^{-\frac{3}{2}}$$

power down, minus one

$$\text{ANSWER } 3x^{\frac{1}{2}} - \frac{9}{4}x^{-\frac{3}{2}}$$

× **TRAP** $\frac{1}{2\sqrt{x}}$ is $\frac{1}{2}x^{-\frac{1}{2}}$, not $2x^{-\frac{1}{2}}$. The number stays on the bottom.

IDEA 01

Practise: Rewrite First, Then Differentiate

My questions first, then selected practice, then real past papers.

Q1 DR ESLAM Roots become powers

P1 · TOPIC 12 3 marks

Differentiate (a) $\sqrt{x}$ (b) $\frac{4}{x^3}$ (c) $\frac{6}{\sqrt[3]{x}}$

Q2 DR ESLAM Split the fraction

P1 · TOPIC 12 3 marks

Find $\frac{dy}{dx}$ when $y = \frac{x^3 - 4x}{\sqrt{x}}$.

Divide each term by $x^{\frac{1}{2}}$ first.

SELECTED PRACTICE

Q3**PRACTICE**

Practice · Easy

P1 · TOPIC 12 3 marks

1 Differentiate

(i) $5x$,

(ii) $2x^3$,

(iii) $x^{\frac{1}{2}}$.

(3 marks)

Q4**PRACTICE**

Practice · Medium

P1 · TOPIC 12 4 marks

11 (a) Given that $y = 2x^3 - 8\sqrt{x}$, find

$$\frac{dy}{dx}$$

(2 marks)

(b) $\frac{d^2y}{dx^2}$

(2 marks)

Q5**PRACTICE**

Practice · Very Hard

P1 · TOPIC 12 3 marks

7 A curve is described by the equation $\frac{\sqrt{y}}{-1 + \sqrt{x}} = \frac{1}{x}$, $x > 1$. Find $\frac{dy}{dx}$.

(3 marks)

PAST PAPERS

Q6 PAST PAPER Pearson IAL WMA11 · January 2023 · Q1

P1 · TOPIC 12 5 marks

1. A curve C has equation

$$y = 2 + 10x^{\frac{1}{2}} - 2x^{\frac{3}{2}} \quad x > 0$$

(a) Find $\frac{dy}{dx}$ giving your answer in simplest form.

(3)

(b) Hence find the exact value of the gradient of the tangent to C at the point where $x = 2$ giving your answer in simplest form.*(Solutions relying on calculator technology are not acceptable.)*

(2)

(Total for Question 1 is 5 marks)

Q7**PAST PAPER**

Pearson IAL WMA11 · June 2023 · Q4

P1 · TOPIC 12 7 marks

4. In this question you must show all stages of your working.

(a) Write

$$y = \frac{5x^2 + \sqrt{x^3}}{\sqrt[3]{8x}}$$

in the form

$$y = Ax^p + Bx^q$$

where A , B , p and q are constants to be found.

(4)

(b) Hence find $\frac{dy}{dx}$ giving each coefficient in simplest form.

(3)

(Total for Question 4 is 7 marks)

Q8 **PAST PAPER** Pearson IAL WMA11 · October 2023 · Q1**P1 · TOPIC 12** **5 marks**

1. Given that

$$y = 5x^3 + \frac{3}{x^2} - 7x \quad x > 0$$

find, in simplest form,

(a) $\frac{dy}{dx}$ (3)

(b) $\frac{d^2y}{dx^2}$ (2)

(Total for Question 1 is 5 marks)

02 Using the Gradient

$\frac{dy}{dx}$ is a gradient machine: put x in, get the gradient out — or set it equal to a number and solve for x .

WHEN YOU SEE → YOU DO

“find the gradient at $x = 2$ ”

→ substitute into $\frac{dy}{dx}$

“the gradient is 4” / “at P and Q the gradient is 2”

→ $\frac{dy}{dx} = 4 \rightarrow$ solve for x

“stationary point”, “ $\frac{dy}{dx} = 0$ ”, “turning point”

→ solve $\frac{dy}{dx} = 0$, then find y

“the gradient is less than -18 ”

→ solve the inequality $\frac{dy}{dx} < -18$

“ a, b, c ” with gradients at two points

→ one equation per point → simultaneous

WORKED MODEL · STATIONARY POINT

$y = 2x^3 - 9x^2 + 12x$ — where is the gradient zero?

1 $\frac{dy}{dx} = 6x^2 - 18x + 12 = 0$

differentiate, = 0

2 $x^2 - 3x + 2 = 0 \Rightarrow x = 1, 2$

solve (calculator first)

ANSWER (1, 5) and (2, 4)

DISGUISED QUADRATIC $12x^4 + 5x^2 - 2 = 0$ or $4x - 5\sqrt{x} - 44 = 0$: let $u = x^2$ or $u = \sqrt{x}$, solve, then go back.

IDEA 02

Practise: Using the Gradient

My questions first, then selected practice, then real past papers.

Q9**DR ESLAM**

Gradient given

P1 · TOPIC 12**3 marks**

Find the point on $y = x^2 - 6x + 1$ where the gradient is 4.

Q10**DR ESLAM**

Stationary points

P1 · TOPIC 12**4 marks**

Find the coordinates of the stationary points of $y = x^3 - 3x + 2$.

SELECTED PRACTICE

Q11**PRACTICE**

Practice · Easy

P1 · TOPIC 12

4 marks

2 (a) Write down the gradient of the line with equation $y = k$, where k is a constant.

(1 mark)

(b) Find the gradient at the point where $x = 8$ for the following functions

(i) $f(x) = 3x^2$,

(ii) $f(x) = 4x^3 - 2x$,

(iii) $f(x) = 3x^{\frac{1}{3}}$.

(3 marks)

Q12**PRACTICE**

Practice · Medium

P1 · TOPIC 12 4 marks

7 (a) A curve is described by the equation $\frac{y}{x-3} = x^2 + 1$.

Make y the subject of the equation.

(1 mark)

(b) Hence find $\frac{dy}{dx}$.

(1 mark)

(c) Find the coordinates of the point on the curve where the gradient is -2 .

(2 marks)

Q13**PRACTICE**

Practice · Very Hard

P1 · TOPIC 12 5 marks

- 8 The curve with equation $y = ax^2 + bx + c$ passes through the point $(-1, 4)$. At the point $(2, 7)$ the gradient of the curve is 7. Find the values of a , b and c .

(5 marks)

PAST PAPERS

Q14 PAST PAPER Pearson IAL WMA11 · January 2020 · Q9

P1 · TOPIC 12 6 marks

9. In this question you must show all stages of your working.

Solutions relying on calculator technology are not acceptable.

A curve has equation

$$y = \frac{4x^2 + 9}{2\sqrt{x}} \quad x > 0$$

Find the x coordinate of the point on the curve at which $\frac{dy}{dx} = 0$

(6)

(Total 6 marks)

Q15 **PAST PAPER** Pearson IAL WMA11 · January 2021 · Q7**P1 · TOPIC 12** **9 marks**

7. In this question you must show all stages of your working.

Solutions relying on calculator technology are not acceptable.

$$f(x) = 2x - 3\sqrt{x} - 5 \quad x > 0$$

(a) Solve the equation

$$f(x) = 9 \quad (4)$$

(b) Solve the equation

$$f''(x) = 6 \quad (5)$$

(Total 9 marks)

Q16**PAST PAPER**

Pearson IAL WMA11 · October 2021 · Q2

P1 · TOPIC 12 5 marks

2. In this question you must show all stages of your working.

Solutions relying on calculator technology are not acceptable.

A curve has equation

$$y = 3x^5 + 4x^3 - x + 5$$

The points P and Q lie on the curve.

The gradient of the curve at both point P and point Q is 2

Find the x coordinates of P and Q .

(5)

(Total 5 marks)

7.

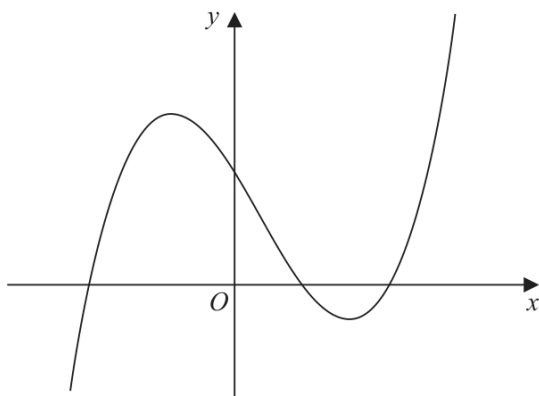


Figure 3

Figure 3 shows a sketch of part of the curve with equation $y = f(x)$, where

$$f(x) = (x + 4)(x - 2)(2x - 9)$$

Given that the curve with equation $y = f(x) - p$ passes through the point with coordinates $(0, 50)$

- (a) find the value of the constant p . (2)

Given that the curve with equation $y = f(x + q)$ passes through the origin,

- (b) write down the possible values of the constant q . (2)

- (c) Find $f'(x)$. (4)

- (d) Hence find the range of values of x for which the gradient of the curve with equation $y = f(x)$ is less than -18 (3)

(Total 11 marks)

Q18 **PAST PAPER** Pearson IAL WMA11 · June 2024 · Q7**P1 · TOPIC 12** **8 marks**

7. The curve C has equation $y = f(x)$ where

$$f(x) = 2x^3 - kx^2 + 14x + 24$$

and k is a constant.

(a) Find, in simplest form,

(i) $f'(x)$

(ii) $f''(x)$

(3)

The curve with equation $y = f'(x)$ intersects the curve with equation $y = f''(x)$ at the points A and B .

Given that the x coordinate of A is 5

(b) find the value of k .

(2)

(c) Hence find the coordinates of B .

(3)

(Total for Question 7 is 8 marks)

Q19**PAST PAPER**

Pearson IAL WMA11 · October 2024 · Q3

P1 · TOPIC 12 9 marks

3. In this question you must show all stages of your working.

$$f(x) = \frac{(x+5)^2}{\sqrt{x}} \quad x > 0$$

(a) Find $\int f(x) \, dx$

(4)

(b) (i) Show that when $f'(x) = 0$

$$3x^2 + 10x - 25 = 0$$

(ii) Hence state the value of x for which

$$f'(x) = 0$$

(5)**(Total for Question 3 is 9 marks)**

9.

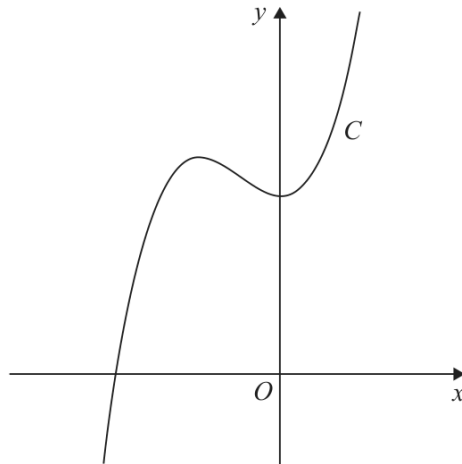
**Figure 4**

Figure 4 shows a sketch of the curve C with equation $y = f(x)$, where

$$f(x) = (x + 5)(3x^2 - 4x + 20)$$

(a) Deduce the range of values of x for which $f(x) \geq 0$ (1)

(b) Find $f'(x)$ giving your answer in simplest form. (3)

The point $R(-4, 84)$ lies on C .

Given that the tangent to C at the point P is parallel to the tangent to C at the point R

(c) find the x coordinate of P . (4)

(d) Find the point to which R is transformed when the curve with equation $y = f(x)$ is transformed to the curve with equation,

- (i) $y = f(x - 3)$
- (ii) $y = 4f(x)$ (2)

(Total for Question 9 is 10 marks)

Q21 **PAST PAPER** Pearson IAL WMA11 · June 2025 · Q2**P1 · TOPIC 12** **5 marks**

2. The curve C has equation

$$y = 2x^{\frac{5}{2}} - 4x + 3$$

- (a) Find $\frac{dy}{dx}$ writing your answer in simplest form.

(2)

The point P lies on C .

Given that

- the x coordinate of P is 2^k where k is a constant
- the gradient of C at the point P is 16

- (b) find the value of k .

(3)

(Total for Question 2 is 5 marks)

Q22**PAST PAPER**

Pearson IAL WMA11 · June 2026 · Q4

P1 · TOPIC 12 10 marks

4. **In this question you must show all stages of your working.**
Solutions relying on calculator technology are not acceptable.

$$f(x) = x^3 - 2x^2\sqrt{x} + 14 \quad x > 0$$

- (a) Solve the equation

$$f'(x) = 0 \quad (5)$$

- (b) (i) Show that the equation

$$f''(x) = 66$$

can be written as

$$4x - 5x^{\frac{1}{2}} - 44 = 0$$

- (ii) Hence, using algebra, show that $x = 16$ is the only solution of the equation

$$f''(x) = 66 \quad (5)$$

Question 4 continued

(Total for Question 4 is 10 marks)

03 Tangents and Normals

The tangent has the gradient of the curve. The normal is perpendicular: flip and change sign.

WHEN YOU SEE → YOU DO

“tangent to C at P ”

$$\rightarrow m = \frac{dy}{dx} \text{ at } P, \text{ then } y - y_1 = m(x - x_1)$$

“normal to C at P ”

$$\rightarrow m = -\frac{1}{dy/dx}$$

“in the form $ax + by + c = 0$ where a, b, c are integers”

→ clear fractions, all on one side

“the tangent/normal meets C again”

→ curve = line, solve (you already know one root)

“parallel to the tangent at R ”

→ same gradient: $\frac{dy}{dx} = \text{that number}$

WORKED MODEL · JUNE 2021

Normal to $y = \frac{x^2}{3} + \frac{4}{\sqrt{x}} + \frac{8}{3x} - 5$ at $P(4, 3)$.

1 $\frac{dy}{dx} = \frac{2}{3}x - 2x^{-\frac{3}{2}} - \frac{8}{3}x^{-2}$

rewrite, differentiate

2 at $x = 4$: $\frac{8}{3} - \frac{1}{4} - \frac{1}{6} = \frac{9}{4}$

gradient of tangent

3 $m_N = -\frac{4}{9}$: $y - 3 = -\frac{4}{9}(x - 4)$

normal

ANSWER $4x + 9y - 43 = 0$

IDEA 03

Practise: Tangents and Normals

My questions first, then selected practice, then real past papers.

Q23 DR ESLAM Tangent

P1 · TOPIC 12 3 marks

Find the equation of the tangent to $y = x^3 - 2x$ at $x = 2$.

Q24 DR ESLAM Normal, integer form

P1 · TOPIC 12 4 marks

Find the normal to $y = \frac{8}{x}$ at $(2, 4)$ in the form $ax + by + c = 0$.

SELECTED PRACTICE

Q25

PRACTICE

Practice · Easy

P1 · TOPIC 12 5 marks

10 (i) Find the gradient of the tangent at the point $(2, 3)$ on the graph of $y = 2x^3 - 3x^2 - 1$.

(ii) Hence find the equation of the tangent at the point $(2, 3)$.

(5 marks)**Q26**

PRACTICE

Practice · Hard

P1 · TOPIC 12 5 marks

9 The curve C has equation $y = 3x^2 - 6x + \sqrt{2x}$. The point $P(2, 2)$ lies on C .

Find an equation of the tangent to C at P .

(5 marks)

Q27**PRACTICE**

Practice · Very Hard

P1 · TOPIC 12 11 marks

10 (a) A curve is described by the equation $y = f(x)$, where

$$f(x) = \frac{1}{\sqrt{x}}, x > 0$$

P is the point on the curve such that the normal to the curve at P also passes through the origin.

Find the coordinates of point P . Give your answer in the form $(2^a, 2^b)$, where a and b are rational numbers to be found.

(6 marks)

(b) Write down the equation of the normal to the curve at P .

(1 mark)

(c) Show that an equation of the tangent to the curve at P is

$$\left(2^{\frac{1}{3}}\right)x + \left(2^{\frac{5}{6}}\right)y = 3$$

(4 marks)

PAST PAPERS

Q28 **PAST PAPER** Pearson IAL WMA11 · October 2019 · Q5**P1 · TOPIC 12** **7 marks**

5. A curve has equation

$$y = \frac{x^3}{6} + 4\sqrt{x} - 15 \quad x \geq 0$$

- (a) Find $\frac{dy}{dx}$, giving the answer in simplest form.

(3)

The point $P\left(4, \frac{11}{3}\right)$ lies on the curve.

- (b) Find the equation of the normal to the curve at P . Write your answer in the form $ax + by + c = 0$, where a , b and c are integers to be found.

(4)**(Total 7 marks)**

Q29 **PAST PAPER** Pearson IAL WMA11 · January 2021 · Q1**P1 · TOPIC 12** **8 marks**

1. A curve has equation

$$y = 2x^3 - 5x^2 - \frac{3}{2x} + 7 \quad x > 0$$

- (a) Find, in simplest form, $\frac{dy}{dx}$ (3)

The point P lies on the curve and has x coordinate $\frac{1}{2}$

- (b) Find an equation of the normal to the curve at P , writing your answer in the form $ax + by + c = 0$, where a , b and c are integers to be found. (5)

(Total 8 marks)

Q30 **PAST PAPER** Pearson IAL WMA11 · June 2021 · Q1**P1 · TOPIC 12** **8 marks**

1. The curve C has equation

$$y = \frac{x^2}{3} + \frac{4}{\sqrt{x}} + \frac{8}{3x} - 5 \quad x > 0$$

- (a) Find $\frac{dy}{dx}$, giving your answer in simplest form.

(4)

The point $P(4, 3)$ lies on C .

- (b) Find the equation of the normal to C at the point P . Write your answer in the form $ax + by + c = 0$, where a , b and c are integers to be found.

(4)**(Total 8 marks)**

Q31 PAST PAPER Pearson IAL WMA11 · June 2022 · Q10

P1 · TOPIC 12 12 marks

10.

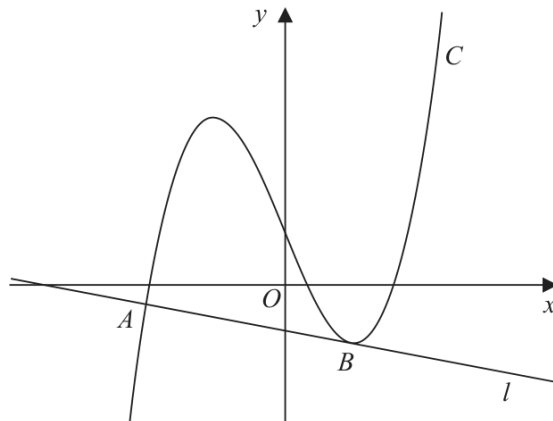
**Figure 5**

Figure 5 shows a sketch of the curve C with equation

$$y = \frac{2}{7}x^3 + \frac{1}{7}x^2 - \frac{5}{2}x + k$$

where k is a constant.

- (a) Find $\frac{dy}{dx}$ (2)

The line l , shown in Figure 5, is the normal to C at the point A with x coordinate $-\frac{7}{2}$

Given that l is also a tangent to C at the point B ,

- (b) show that the x coordinate of the point B is a solution of the equation

$$12x^2 + 4x - 33 = 0 \quad (4)$$

- (c) Hence find the x coordinate of B , justifying your answer. (2)

Given that the y intercept of l is -1

- (d) find the value of k . (4)

(Total 12 marks)

Q32**PAST PAPER**

Pearson IAL WMA11 · October 2022 · Q1

P1 · TOPIC 12 6 marks

1. The curve C has equation

$$y = \frac{x^3}{4} - x^2 + \frac{17}{x} \quad x > 0$$

- (a) Find $\frac{dy}{dx}$, giving your answer in simplest form.

(3)

The point $R\left(2, \frac{13}{2}\right)$ lies on C .

- (b) Find the equation of the tangent to C at the point R . Write your answer in the form $ax + by + c = 0$, where a , b and c are integers to be found.

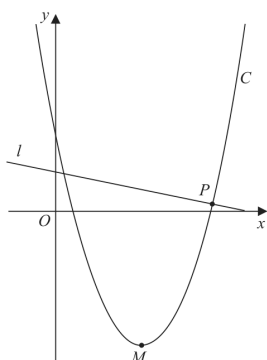
(3)**(Total for Question 1 is 6 marks)**

Q33**PAST PAPER**

Pearson IAL WMA11 · October 2022 · Q9

P1 · TOPIC 12 14 marks

9.

**Figure 3**Figure 3 shows a sketch of the curve C with equation

$$y = \frac{1}{2}x^2 - 10x + 22$$

- (a) Write $\frac{1}{2}x^2 - 10x + 22$ in the form

$$a(x + b)^2 + c$$

where a , b and c are constants to be found.

(3)

The point M is the minimum turning point of C , as shown in Figure 3.

- (b) Deduce the coordinates of M

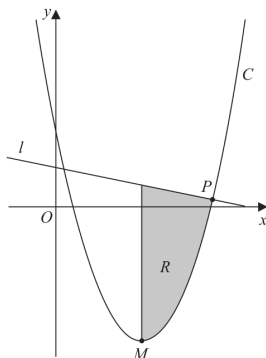
(2)

The line l is the normal to C at the point P , as shown in Figure 3.Given that l has equation $y = k - \frac{1}{8}x$, where k is a constant,

- (c) (i) find the coordinates of P

- (ii) find the value of k

(6)

Question 9 continues on the next page**Figure 4**Figure 4 is a copy of Figure 3. The finite region R , shown shaded in Figure 4, is bounded by l , C and the line through M parallel to the y -axis.

- (d) Identify the inequalities that define R .

(3)

(Total for Question 9 is 14 marks)

Q34 PAST PAPER Pearson IAL WMA11 · January 2023 · Q10

P1 · TOPIC 12 10 marks

10.

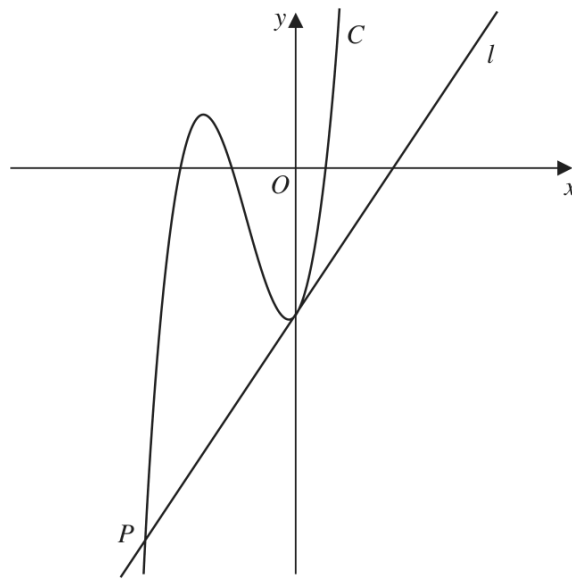
**Figure 4**

Figure 4 shows a sketch of part of the curve C with equation $y = f(x)$, where

$$f(x) = (3x + 20)(x + 6)(2x - 3)$$

- (a) Use the given information to state the values of x for which

$$f(x) > 0$$

(2)

- (b) Expand $(3x + 20)(x + 6)(2x - 3)$, writing your answer as a polynomial in simplest form.

(3)

The straight line l is the tangent to C at the point where C cuts the y -axis.

Given that l cuts C at the point P , as shown in Figure 4,

- (c) find, using algebra, the x coordinate of P

(Solutions based on calculator technology are not acceptable.)

(5)

(Total for Question 10 is 10 marks)

Q35 **PAST PAPER** Pearson IAL WMA11 · January 2024 · Q3**P1 · TOPIC 12** **7 marks**

3. The curve C has equation

$$y = \frac{5x^3 - 8}{2x^2} \quad x > 0$$

- (a) Find $\frac{dy}{dx}$ writing your answer in simplest form.

(4)

The point $P(2, 4)$ lies on C .

- (b) Find an equation for the tangent to C at P writing your answer in the form $ax + by + c = 0$, where a , b and c are integers.

(3)**(Total for Question 3 is 7 marks)**

Q36 **PAST PAPER** Pearson IAL WMA11 · January 2025 · Q5**P1 · TOPIC 12** **8 marks**

5. **In this question you must show all stages of your working.
Solutions relying on calculator technology are not acceptable.**

The curve C has equation

$$y = 4x^3 + \frac{2}{x} + 9 \quad x > 0$$

- (a) Find $\frac{dy}{dx}$, giving your answer in simplest form.

(2)

Given that

- the point P lies on C
- the line with equation $y = k - 5x$, where k is a constant, is the tangent to C at P

- (b) show that the x coordinate of P satisfies the equation

$$12x^4 + 5x^2 - 2 = 0$$

(2)

- (c) Hence find the value of k .

(4)**(Total for Question 5 is 8 marks)**

Q37 PAST PAPER Pearson IAL WMA11 · October 2025 · Q1

P1 · TOPIC 12 6 marks

1. The curve C has equation

$$y = 6x^2 + 3\sqrt{x} + \frac{5}{8} \quad x > 0$$

- (a) Find $\frac{dy}{dx}$ writing the answer in simplest form.

(3)

The point $P\left(\frac{1}{4}, \frac{5}{2}\right)$ lies on C .

- (b) Find the equation of the tangent to C at P , writing your answer in the form $y = mx + c$ where m and c are integers.

(3)

(Total for Question 1 is 6 marks)

10.

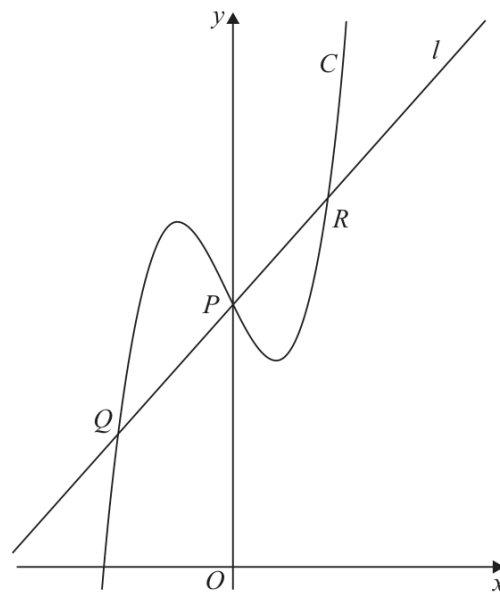


Figure 4

**In this question you must show all stages of your working.
Solutions relying entirely on calculator technology are not acceptable.**

Figure 4 shows a sketch of the curve C with equation

$$y = 2x^3 + \frac{1}{2}x^2 - 2x + 5$$

The line l is the normal to C at the point P where $x = 0$

The line l also intersects C at points Q and R as shown in Figure 4.

(a) Find, using algebra, the x coordinate of point Q .

(6)

The point T lies on C .

Given that

- the tangent to C at T is parallel to l
- the x coordinate of T is positive

(b) find, using algebra, the exact x coordinate of T .

(4)

(Total for Question 10 is 10 marks)

04 How Many Stationary Points? The Discriminant

$f'(x)$ of a cubic is a quadratic. Its discriminant tells you how many points have zero gradient.

WHEN YOU SEE → YOU DO

“exactly one point where the gradient is zero”

→ $b^2 - 4ac = 0$ for $f'(x)$

“no solutions to $f'(x) = 0$ ”

→ $b^2 - 4ac < 0$

“at least one real solution”

→ $b^2 - 4ac \geq 0$

“ $\frac{dy}{dx}$ in the form $p(x + q)^2 + r$ ”

→ complete the square: min gradient = r at $x = -q$

WORKED MODEL · NO STATIONARY POINTS

$$f(x) = x^3 - 4x^2 + 6x - 9$$

1 $f'(x) = 3x^2 - 8x + 6$

differentiate

2 $b^2 - 4ac = 64 - 72 = -8 < 0$

discriminant

ANSWER $f'(x) = 0$ has no real solutions

IDEA 04

Practise: How Many Stationary Points?

My questions first, then selected practice, then real past papers.

Q39 **DR ESLAM** Discriminant of $f'(x)$

P1 · TOPIC 12 3 marks

$f(x) = x^3 + kx^2 + 3x$. Find the values of k for which $f'(x) = 0$ has exactly one solution.

SELECTED PRACTICE**Q40** **PRACTICE** Practice · Medium

P1 · TOPIC 12 4 marks

6 (a) The function f is defined by $f(x) = x^3 - 6x^2 - cx + 12$.

Find $f'(x)$.

(2 marks)

(b) Given that the equation $f'(x) = 0$ has exactly one real solution, find the value of c .

(2 marks)

Q41**PRACTICE**

Practice · Hard

P1 · TOPIC 12 4 marks

- 6 A curve has the equation $y = 4x^3 + bx^2 + 3x - 17$, where b is a constant. Given that there is only one point on the curve where the gradient is zero, determine the possible values of b .

(4 marks)**Q42****PRACTICE**

Practice · Very Hard

P1 · TOPIC 12 4 marks

- 6 The function f is defined by $f(x) = x^n - x$, $n \in \mathbb{N}$, $n \geq 2$. Determine the relationship between the value of n and the number of real solutions to the equation $f'(x) = 0$.

(4 marks)

PAST PAPERS

Q43 PAST PAPER Pearson IAL WMA11 · January 2026 · Q7

P1 · TOPIC 12 10 marks

7. The curve C has equation

$$y = \frac{2}{3}x^3 - 8x^2 + 43x - \frac{20}{3}$$

(a) Show that $\frac{dy}{dx}$ can be written in the form

$$p(x + q)^2 + r$$

where p , q and r are constants to be found.

(5)

(b) Hence state

(i) the minimum value of $\frac{dy}{dx}$ (ii) the value of x at which this minimum value occurs.

(2)

Given that S is the point on C at which the gradient is a minimum,(c) find the equation of the tangent to C at S , giving your answer in the form $y = mx + c$, where m and c are constants.

(3)

(Total for Question 7 is 10 marks)

05 From First Principles: the Chord PQ

The gradient of a chord from P to a point h away tends to the gradient of the tangent as $h \rightarrow 0$.

WHEN YOU SEE $\rightarrow$ YOU DO

“Q with x coordinate $2 + h$ ”

$$\rightarrow \text{gradient } PQ = \frac{y_Q - y_P}{h}$$

“explain the relationship”

$\rightarrow$ as $h \rightarrow 0$, the chord gradient $\rightarrow$ the tangent gradient

WORKED MODEL · JANUARY 2020

$y = x^2 + 3x - 2$, $P(3, 16)$, Q at $x = 3 + h$.

1 $y_Q = (3 + h)^2 + 3(3 + h) - 2 = 16 + 9h + h^2$

substitute

2 gradient = $\frac{9h + h^2}{h}$

change in y over h

ANSWER $9 + h \rightarrow 9$ as $h \rightarrow 0$

IDEA 05

Practise: From First Principles

My questions first, then real past papers.

Q44 **DR ESLAM** Chord to tangent

P1 · TOPIC 12 3 marks

$y = x^2$. P is $(1, 1)$ and Q has x coordinate $1 + h$. Find the gradient of PQ and its limit as $h \rightarrow 0$.

PAST PAPERS

Q45 PAST PAPER Pearson IAL WMA11 · October 2019 · Q7

P1 · TOPIC 12 6 marks

7.

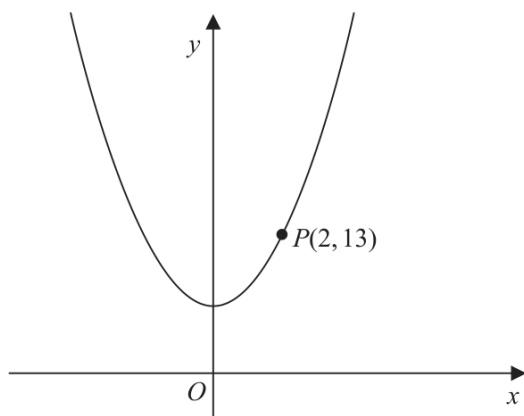
**Figure 4**

Figure 4 shows part of the curve with equation $y = 2x^2 + 5$

The point $P(2, 13)$ lies on the curve.

- (a) Find the gradient of the tangent to the curve at P . (2)

The point Q with x coordinate $2 + h$ also lies on the curve.

- (b) Find, in terms of h , the gradient of the line PQ . Give your answer in simplest form. (3)

- (c) Explain briefly the relationship between the answer to (b) and the answer to (a). (1)

(Total 6 marks)

Q46**PAST PAPER**

Pearson IAL WMA11 · January 2020 · Q3

P1 · TOPIC 12 6 marks

3.

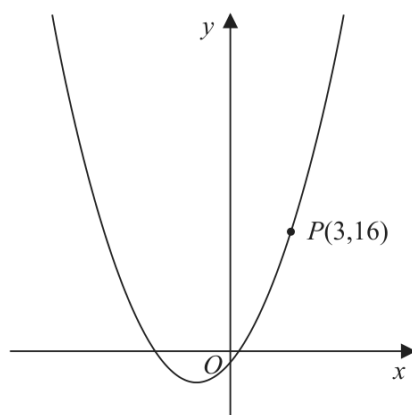
**Figure 1**

Figure 1 shows part of the curve with equation $y = x^2 + 3x - 2$

The point $P(3,16)$ lies on the curve.

(a) Find the gradient of the tangent to the curve at P .

(2)

The point Q with x coordinate $3 + h$ also lies on the curve.

(b) Find, in terms of h , the gradient of the line PQ . Write your answer in simplest form.

(3)

(c) Explain briefly the relationship between the answer to (b) and the answer to (a).

(1)**(Total 6 marks)**

ALL The Whole Topic on One Page

Tear this page out before the exam.

- 01 Rewrite** Roots → fractional powers, “over” → negative powers, brackets → expand.
- 02 Use it** Gradient at x : substitute. Given gradient: $\frac{dy}{dx} = k$, solve.
- 03 Tangent / normal** $m_T = \frac{dy}{dx}$, $m_N = -\frac{1}{m_T}$, $y - y_1 = m(x - x_1)$.
- 04 How many?** Discriminant of $f'(x)$. Min gradient: complete the square.
- 05 First principles** $\frac{f(x+h) - f(x)}{h} \rightarrow f'(x)$.

BOTTOM LINE Rewrite → differentiate → substitute or solve → “tangent” means the derivative, “normal” means $-\frac{1}{\text{derivative}}$.

Answers

Differentiation • for the teacher. Every answer checked with a computer algebra system.

IDEA 01 • REWRITE FIRST, THEN DIFFERENTIATE

- Q1** Dr Eslam • Roots become powers
(a) $\frac{1}{2}x^{-\frac{1}{2}}$ (b) $-12x^{-4}$ (c) $-2x^{-\frac{4}{3}}$
- Q2** Dr Eslam • Split the fraction
 $\frac{5}{2}x^{\frac{3}{2}} - 2x^{-\frac{1}{2}}$
- Q3** Practice • Easy
(i) 5 (ii) $6x^2$ (iii) $\frac{1}{2}x^{-\frac{1}{2}}$
- Q4** Practice • Medium
(a) $6x^2 - 4x^{-\frac{1}{2}}$ (b) $12x + 2x^{-\frac{3}{2}}$
- Q5** Practice • Very Hard
 $-x^{-2} + 3x^{-\frac{5}{2}} - 2x^{-3}$
- Q6** Pearson IAL WMA11 • January 2023 • Q1
(a) $5x^{-\frac{1}{2}} - 3x^{\frac{1}{2}}$ (b) $-\frac{\sqrt{2}}{2}$
- Q7** Pearson IAL WMA11 • June 2023 • Q4
(a) $\frac{5}{2}x^{\frac{5}{3}} + \frac{1}{2}x^{\frac{7}{6}}$ (b) $\frac{25}{6}x^{\frac{2}{3}} + \frac{7}{12}x^{\frac{1}{6}}$
- Q8** Pearson IAL WMA11 • October 2023 • Q1
(a) $15x^2 - 6x^{-3} - 7$ (b) $30x + 18x^{-4}$

IDEA 02 • USING THE GRADIENT

- Q9** Dr Eslam · Gradient given $2x - 6 = 4 \Rightarrow (5, -4)$
- Q10** Dr Eslam · Stationary points $(1, 0)$ and $(-1, 4)$
- Q11** Practice · Easy (a) 0 (b) 48, 766, $\frac{1}{4}$
- Q12** Practice · Medium (a) $y = x^3 - 3x^2 + x - 3$ (b) $3x^2 - 6x + 1$ (c) $(1, -4)$
- Q13** Practice · Very Hard $a = 2, b = -1, c = 1$
- Q14** Pearson IAL WMA11 · January 2020 · Q9 $x = \frac{\sqrt{3}}{2}$
- Q15** Pearson IAL WMA11 · January 2021 · Q7 (a) $x = \frac{49}{4}$ (b) $x = \frac{1}{4}$
- Q16** Pearson IAL WMA11 · October 2021 · Q2 $x = \pm \frac{1}{\sqrt{5}}$
- Q17** Pearson IAL WMA11 · January 2022 · Q7 (a) $p = 22$ (b) $q = -4, 2, 4.5$ (c) $6x^2 - 10x - 34$
(d) $-1 < x < \frac{8}{3}$
- Q18** Pearson IAL WMA11 · June 2024 · Q7 (a) $6x^2 - 2kx + 14, 12x - 2k$ (b) $k = 13$ (c) $B(\frac{4}{3}, -10)$
- Q19** Pearson IAL WMA11 · October 2024 · Q3 (a) $\frac{2}{5}x^{\frac{5}{2}} + \frac{20}{3}x^{\frac{3}{2}} + 50x^{\frac{1}{2}} + c$ (b)(ii) $x = \frac{5}{3}$
- Q20** Pearson IAL WMA11 · October 2024 · Q9 (a) $x \geq -5$ (b) $9x^2 + 22x$ (c) $x = \frac{14}{9}$ (d) $(-1, 84), (-4, 336)$
- Q21** Pearson IAL WMA11 · June 2025 · Q2 (a) $5x^{\frac{3}{2}} - 4$ (b) $k = \frac{4}{3}$
- Q22** Pearson IAL WMA11 · June 2026 · Q4 (a) $x = \frac{25}{9}$ (b)(ii) $\sqrt{x} = 4$ ($\sqrt{x} = -\frac{11}{4}$ rejected)

IDEA 03 · TANGENTS AND NORMALS

- Q23** Dr Eslam · Tangent $y = 10x - 16$
- Q24** Dr Eslam · Normal, integer form gradient -2, normal gradient $\frac{1}{2}$: $x - 2y + 6 = 0$
- Q25** Practice · Easy (i) 12 (ii) $y = 12x - 21$
- Q26** Practice · Hard $y = \frac{13}{2}x - 11$
- Q27** Practice · Very Hard (a) $P(2^{-\frac{1}{3}}, 2^{\frac{1}{6}})$ (b) $y = \sqrt{2}x$
- Q28** Pearson IAL WMA11 · October 2019 · Q5 (a) $\frac{1}{2}x^2 + 2x^{-\frac{1}{2}}$ (b) $x + 9y - 37 = 0$
- Q29** Pearson IAL WMA11 · January 2021 · Q1 (a) $6x^2 - 10x + \frac{3}{2}x^{-2}$ (b) $2x + 5y - 16 = 0$
- Q30** Pearson IAL WMA11 · June 2021 · Q1 (a) $\frac{2}{3}x - 2x^{-\frac{3}{2}} - \frac{8}{3}x^{-2}$ (b) $4x + 9y - 43 = 0$
- Q31** Pearson IAL WMA11 · June 2022 · Q10 (a) $\frac{6}{7}x^2 + \frac{2}{7}x - \frac{5}{2}$ (c) $x = \frac{3}{2}$ (d) $k = \frac{5}{4}$
- Q32** Pearson IAL WMA11 · October 2022 · Q1 (a) $\frac{3}{4}x^2 - 2x - 17x^{-2}$ (b) $21x + 4y - 68 = 0$
- Q33** Pearson IAL WMA11 · October 2022 · Q9 (a) $\frac{1}{2}(x - 10)^2 - 28$ (b) $M(10, -28)$ (c) $P(18, 4)$, $k = \frac{25}{4}$ (d) $10 \leq x \leq 18$, curve $\leq y \leq$ line
- Q34** Pearson IAL WMA11 · January 2023 · Q10 (a) $-\frac{20}{3} < x < -6$ or $x > \frac{3}{2}$ (b) $6x^3 + 67x^2 + 126x - 360$ (c) $x = -\frac{67}{6}$
- Q35** Pearson IAL WMA11 · January 2024 · Q3 (a) $\frac{5}{2} + 8x^{-3}$ (b) $7x - 2y - 6 = 0$
- Q36** Pearson IAL WMA11 · January 2025 · Q5 (a) $12x^2 - 2x^{-2}$ (c) $x = \frac{1}{2}$, $k = 16$
- Q37** Pearson IAL WMA11 · October 2025 · Q1 (a) $12x + \frac{3}{2}x^{-\frac{1}{2}}$ (b) $y = 6x + 1$
- Q38** Pearson IAL WMA11 · October 2025 · Q10 (a) $x = -\frac{5}{4}$ (b) $x = \frac{-1 + \sqrt{61}}{12}$

IDEA 04 · HOW MANY STATIONARY POINTS?

- Q39** Dr Eslam · Discriminant of $f'(x)$ $4k^2 - 36 = 0 \Rightarrow k = \pm 3$
- Q40** Practice · Medium (a) $3x^2 - 12x - c$ (b) $c = -12$
- Q41** Practice · Hard $b = \pm 6$
- Q42** Practice · Very Hard n even: one solution · n odd: two solutions
- Q43** Pearson IAL WMA11 · January 2026 · Q7 (a) $2(x-4)^2 + 11$ (b) 11 at $x = 4$ (c) $y = 11x + 36$

IDEA 05 · FROM FIRST PRINCIPLES

- Q44** Dr Eslam · Chord to tangent $2 + h \rightarrow 2$
- Q45** Pearson IAL WMA11 · October 2019 · Q7 (a) 8 (b) $8+2h$ (c) as $h \rightarrow 0$ the chord gradient $\rightarrow 8$
- Q46** Pearson IAL WMA11 · January 2020 · Q3 (a) 9 (b) $9 + h$ (c) as $h \rightarrow 0$, $9 + h \rightarrow 9$